

CATIA SYLLABUS		
OBJECTIVE: The objective of this training Session is to impart knowledge for the students about Catia V5 on Solid fundamentals of the CAD tool to prepare the student for more specific and advanced functions		
Duration:		
S.NO	CONTENT	HOURS (L/P)
1	Introduction to Catia V5 Usage File Operations	
	Workbenches	
	Toolbars	
	Specification Tree	
	Compass	
	Mouse	
2	Drawing Sketches in the Sketcher Workbench-I	
	Invoking the Sketcher Workbench	
	Units, Grid Settings	
	Construction and Standard elements	
	Drawing Sketch objects I	
	Inferred Constraints	
	Exiting the Sketcher Workbench	
	View Manipulation	
3	Drawing Sketches in the Sketcher Workbench-II	
	Drawing Sketch Objects II	
	Modifying sketch Objects	
	Sketch Transformations	
4	Constraining Sketches and Creating Base Features	
	Constraining Sketches	
	Geometric Constraints	
	Dimensional Constraints	
	Sketch Analysis	
	Creating an Extruded Solid Using the Pad Feature	
	Creating a Revolved Solid Using the Shaft Feature	
	Assigning a Material to the Model	
5	Reference Elements and Sketch-Based Features	
	Reference Elements	
	Drafted Filleted Pad Features	
	Multi-Pad Features	
	Pocket Features	
	Drafted Filleted Pocket	
	Multi-Pocket Features	
	Groove	
	Extruding and Revolving Faces	
	Projecting 3D Elements	
6	Creating Dress-Up and Hole Features	
	Hole features	
	Chamfer	
	Fillets	
	Draft	
	Shell	

7	Editing Features	
	Cut, Copy and Paste	
	Deleting Features	
	Deactivating Features	
	Defining the work Object	
	Reordering Features	
	Parent Child Relationships	
	Update Diagnostics	
	Measuring Elements	
8	Transformation Features and Advanced Modeling Tools - I	
	Translating and Rotating Bodies	
	Symmetry and mirror Features	
	Patterns	
	Scale	
	Parts with Multiple Bodies	
	Boolean Operations	
	Stiffener Features	
9	Advanced Modelling Tools-II	
	Rib	
	Slot	
	Multi-section Solids	
	Power copy	
10	Wireframe and surface Design	
	Surfacing Workbenches	
	Wireframe Geometry	
	Extruded Surfaces	
	Revolved Surface	
	Offset Surfaces	
	Swept Surfaces	
	Fill Surface	
	Multi section Surfaces	
	Blended Surfaces	
	Split, Trim and Join	
11	Project Work - 1 (Students Choice)	
	Project Work - 1 (Commercial Project)	